

From the Yakhchal to the Two-Door Refrigerator: Historical Foundations, Market Dynamics, and an Incremental Innovation Strategy for the Domestic Refrigeration Industry

A Product Strategy and Innovation Management Study

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Abstract

This paper examines the domestic refrigerator as a case study in incremental product innovation, tracing the technology from ancient Persian yakhchal ice houses to the contemporary two-door domestic unit as evidence of a long-matured technological trajectory. Drawing on a mixed-methods market research design that combines field surveys, e-commerce platform data, and expert interviews, the study characterizes household demand, brand preference, and purchasing behavior in the domestic market, and situates these findings within a targeted analysis of component architecture, manufacturing process, and quality/energy certification protocols. Applying the technology- market innovation matrix and the Ansoff growth matrix, the study argues that, under current industry conditions, the appropriate innovation and product strategy for the domestic refrigerator is not radical innovation or entry into new markets, but continuous, incremental improvement within existing technology and an existing market, a positioning with direct implications for how domestic manufacturers should compete against established foreign brands and the informal grey market(Mehdipour & Namazian, 2012; scott & walker,).

Keywords: *Domestic refrigerators; Product strategy; Incremental innovation; Technology maturity; Technology-market innovation matrix; Ansoff matrix; Competitive positioning; Two-door refrigerator segment*

1. Introduction

1.1 Historical Background and the Maturation of Refrigeration Technology

Since antiquity, humans have employed various natural cooling and storage methods to preserve food and delay spoilage, including springs, wells, subterranean caves, water reservoirs (Ab-Anbars), and ice houses (Yakhchals). These practices represent some of the earliest efforts to regulate temperature for the preservation of perishable goods. Ancient Persians were among the pioneers of such practices, developing sophisticated ice-storage structures that integrated shallow ice-freezing pools, tall shading walls, and domed, insulated reservoirs. By harnessing natural cooling mechanisms, particularly evaporative and radiative cooling, these structures enabled the production and long-term storage of ice throughout the summer months(Mehdipour & Namazian, 2012).

The sophisticated design of these structures, including effective insulation, solar-radiation management, and the integration of windcatchers (Badgirs) and Qanat water

channels, demonstrates the advanced understanding of natural cooling principles in pre-modern societies. In Iran and other regions, such technologies represent a significant stage in the development of cooling systems, long preceding the emergence of mechanical refrigeration.

The transition from passive to active cooling emerged through a prolonged and cumulative process of technological development rather than from a single transformative invention. In 1756, William Cullen demonstrated the first artificial refrigeration effect. Building on subsequent advances, Oliver Evans proposed a closed-cycle vapor-compression system in 1805, which Jacob Perkins later developed into the first practical vapor-compression refrigerator in 1834 (American Society of Mechanical Engineers, 2020). John Gorrie further advanced mechanical refrigeration with his mechanical cooling unit in 1841, followed by the development and patenting of a commercial ice-making machine in 1853, marking a significant step toward the industrialization of refrigeration technology. This extended sequence of cumulative improvements and capability development, rather than a single radical breakthrough, provides an early example of

the incremental evolution of technology, a pattern that is subsequently examined in the context of the modern domestic refrigerator industry (Hertzmann, 2016).



Figure 1. Historical evolution of refrigeration technology from natural cooling to modern efficient and smart refrigeration systems.

1.2 From Industrial Application to a Mature Domestic Appliance

The introduction of refrigeration technology into domestic settings, beginning in the early twentieth century, represented an extension and adaptation of existing technological capabilities rather than a fundamental reinvention of the technology. In 1914, Nathaniel B. Wales proposed an electric refrigerator concept that contributed to the subsequent development of commercial Kelvinator models (Hertzmann, 2016), while Alfred Mellowes' development of a self-contained compressor unit in 1916 marked an important step toward modern commercial refrigerator manufacturing (American Society of Mechanical Engineers [ASME], 2020). During the 1920s and 1930s, several competing technological architectures coexisted, including the absorption refrigeration system developed by Baltzar von Platen and Carl Munters and subsequently commercialized internationally by Electrolux (Hertzmann, 2016; Scott & Walker, 2026). These systems competed alongside self-contained electric refrigerators, porcelain-enamelled cabinets, and increasingly safer refrigerants such as Freon (Hertzmann, 2016; Scott & Walker, 2026). Over time, however, the self-contained electric refrigerator based on vapor-compression technology gradually emerged as the dominant design, establishing the fundamental architectural configuration that continues to underpin the contemporary domestic refrigerator (Scott & Walker, 2026).

This development is consistent with Utterback and Abernathy's (1975) dynamic model of product and process

innovation, which suggests that the emergence of a dominant design is typically followed by a shift in the nature of technological competition (Scott & Walker, 2026). As the basic product architecture became increasingly standardized, competitive efforts moved away from experimentation with alternative product configurations toward incremental improvements in areas such as energy efficiency, safety, reliability, and environmental performance (Nege & Werke, 2024; Scott & Walker, 2026). This pattern became particularly evident during the 1950s and 1960s and continues to characterize innovation in the domestic refrigerator industry. Recent research in innovation management further suggests that established dominant designs can continue to shape competitive dynamics in the digital era, where user interaction and digital capabilities reinforce and refine the prevailing design rather than fundamentally displace it (Colfer & Baldwin, 2010; Xing & Liu, 2025).

The core function of a refrigerator, namely maintaining a stable internal temperature through the integration of cooling mechanisms, thermal insulation, and heat-transfer processes, extends well beyond the preservation of household food. Refrigeration is also essential for the storage and preservation of pharmaceuticals, vaccines, and other temperature-sensitive products, contributing to the widespread adoption of refrigeration appliances across residential, commercial, and industrial settings (American Society of Mechanical Engineers [ASME], 2020; U.S. Environmental Protection Agency [EPA], 2014). This broad and persistent demand, combined with the relative technological stability of the refrigerator's fundamental architecture, creates a distinctive competitive environment (Langlois & Robertson, 1992; Scott & Walker, 2026). In such a mature market, where the underlying technology itself provides increasingly limited opportunities for differentiation, the central strategic question addressed in Section 2 is how domestic manufacturers can establish and sustain competitive advantage through alternative dimensions of product and innovation strategy (Nege & Werke, 2024).

2. Problem Statement

The domestic refrigerator industry operates within a distinctive competitive environment shaped by the maturity of its core technology, the essential nature of household demand, and the increasing difficulty of achieving meaningful differentiation through conventional technological improvements alone (Nege & Werke, 2024; Scott & Walker, 2026) [524٦٤٨٠]. As refrigeration

technology has become standardized and widely accessible, the fundamental functions of a refrigerator have become increasingly similar across manufacturers, leading to product homogenization (Scott & Walker, 2026; Xing & Liu, 2025) [313^o^ε ∙]. Consequently, competitive advantage is progressively determined by how effectively firms translate evolving customer expectations and market conditions into differentiated product and innovation strategies rather than by the mere adoption of the underlying refrigeration technology (Nege & Werke, 2024).

This challenge is particularly pronounced for domestic manufacturers competing against established foreign brands. Consumer evidence indicates a persistent preference for foreign refrigerator brands, largely associated with perceptions of superior quality, brand reputation, design, and functional performance. In contrast, domestic brands are often selected primarily on the basis of economic considerations, particularly price competitiveness and the availability of local after-sales services. This asymmetry creates a strategic dilemma: while domestic manufacturers possess established production capabilities and can compete through price and service, they face greater difficulty in converting these capabilities into product differentiation that can alter consumer preferences. From the perspective of competitive strategy, this situation is consistent with Teece's (1986) argument that, in mature and widely imitable technological environments, the ability to capture value increasingly depends on complementary assets and the manner in which technological capabilities are commercialized (Colfer & Baldwin, 2010; Langlois & Robertson, 1992; Scott & Walker, 2026) [187^oo^ε ∙, εεε ∙]. Thus, the central challenge is no longer simply how to manufacture a refrigerator, but how to determine which product and innovation directions can generate meaningful market impact under existing technological and competitive constraints.

3. Technology and Product Architecture: Evidence of a Mature Platform

The material composition and component architecture of the domestic refrigerator are examined in this section not as an engineering specification per se, but as empirical evidence supporting the technology-maturity argument developed in Section 6. The high degree of standardization evident in the refrigerator's core materials, functional components, structural configuration, and dimensional parameters suggests that its fundamental technological architecture has reached a relatively mature stage (Langlois & Robertson,

1992; Nege & Werke, 2024; Scott & Walker, 2026). As the core architecture becomes increasingly established, opportunities for radical innovation at the system level become more constrained, while greater potential emerges for incremental improvements in product features, performance, efficiency, usability, and user experience (Nege & Werke, 2024; Scott & Walker, 2026).

Metal remains the principal structural material used in the refrigerator cabinet and outer body, primarily because of its mechanical strength, durability, and suitability for large-scale manufacturing (Hertzmann, 2016; Scott & Walker, 2026). The widespread use of iron and steel in the chassis and outer shell contributes substantially to the characteristic weight and structural rigidity of the appliance. Depending on the product configuration and design requirements, the external surfaces may incorporate aluminum or stainless steel, providing enhanced corrosion resistance while also contributing to the product's aesthetic and perceived quality.

The interior architecture similarly relies on a combination of established material technologies, including metal components, tempered glass, and injection-molded polymers (Xing & Liu, 2025). These materials provide an appropriate balance of durability, relatively low weight, ease of cleaning, manufacturability, and functional flexibility. Between the inner liner and the outer cabinet, polyurethane foam insulation functions as a critical thermal barrier by reducing heat transfer into the refrigerated compartment. At the same time, the insulation layer contributes to the structural integrity of the cabinet by bonding and reinforcing the inner and outer shells. Together, these established material and component choices illustrate the extent to which the fundamental product architecture has become standardized. Table 1 summarizes the principal material and component characteristics together with representative dimensional standards and energy-consumption benchmarks, providing further evidence of the mature technological platform on which contemporary domestic refrigerators are developed (U.S. Environmental Protection Agency [EPA], 2014).

Table 1. Key structural and dimensional specifications of the domestic refrigerator.

Parameter	Material / Specification	Function
Thermal Insulation	Polyurethane foam (thin injected layer between cabinet walls)	Prevents heat ingress, minimizes cold air loss, and reduces energy consumption.
Cabinet Outer Shell	Iron and steel combination (high structural integrity)	Provides mechanical strength and durability.
Cabinet Outer Frame	Aluminum or stainless steel (corrosion-resistant layer)	Enhances durability, protects against moisture, and provides aesthetic appeal.
Interior Liners & Components	Metal and polymer (plastic) composite	Provides lightweight structure, ease of cleaning, and hygienic storage.
Shelves and Drawers	Wire-mesh metal or tempered glass with plastic borders/trims	Offers high load-bearing capacity and prevents spills.
Dimensions (Compact – 3 cu. ft.)	40 × 46 × 57 cm (W × D × H)	Designed for space-constrained environments.
Dimensions (Large – 9 cu. ft.)	54 × 58 × 108 cm (W × D × H)	Maximum standard size for mid-range single-door units.
Standard Unit Dimensions & Weight	92 × 91.6 × 181.6 cm (W × D × H); 115 kg	Standard dimensions for modern domestic refrigerators.
Side-by-Side Refrigerator Energy Use	250–470 kWh/year	High capacity, moderate energy consumption.
Medium Refrigerator Energy Use	≈170 kWh/year	Optimal energy efficiency for smaller households.
Separate Refrigerator-Freezer Energy Use	170–360 kWh/year	Varies based on compressor efficiency and insulation quality.

3.1 Component Architecture

Table 2 summarizes the principal internal and external components of a domestic refrigerator. The high degree of standardization observed across manufacturers, particularly in core components such as the compressor, condenser, evaporator, capillary tube, thermostat, door gasket, and associated parts, provides further evidence of a well-established component architecture. Although manufacturers may vary these components in terms of specifications, performance characteristics, or configuration,

the underlying functional structure remains largely consistent across products. This stability indicates that the core component architecture is no longer undergoing substantial technical transformation, but instead provides an established platform for incremental improvements in performance, efficiency, reliability, and functionality. Accordingly, the standardized nature of the component structure further supports the technology-maturity argument developed in Section 6.

Table 2.. Refrigerator components and their functions

Category	Component	Function
Internal	Compressor	The heart of the refrigeration system; compresses and circulates the refrigerant.
Internal	Condenser	External coiled tubes that release heat from the refrigerant to the surrounding air.
Internal	Evaporator	A heat exchanger inside the refrigerator that absorbs heat from the stored food.
Internal	Capillary Tube	Creates a pressure drop to reduce the boiling point of the refrigerant.
Internal	Filter Drier	Removes moisture and impurities from the refrigerant to protect the system.
Internal	Refrigerant	The cooling fluid that transfers heat from inside the refrigerator to the outside.
Internal	Thermostat	Automatically controls the internal temperature of the refrigerator.
Internal	Diffuser / Air Damper	Regulates the flow of cold air between the freezer and refrigerator compartments.
Internal	Defrost Heater	An electric heater used to melt frost from the evaporator.
Internal	Accumulator	Stores refrigerant and prevents liquid from entering the compressor.
Internal	Overload Protector	Protects the compressor from overheating and excessive current.
External	Refrigerator Compartment	Storage space for food, kept above freezing temperature.
External	Freezer Compartment	Storage space for frozen food, maintained below 0°C.
External	Temperature Control	Used to adjust the refrigerator temperature; may be a knob or digital panel.
External	Light Switch	Turns the refrigerator light on when the door opens and off when it closes.
External	Door Gasket	A flexible rubber seal that prevents warm air and moisture from entering.
External	Shelves	Strong glass or metal surfaces used to store food items.
External	Drain Pan	Collects defrost water and allows it to evaporate near the compressor.
External	Water Tank / Water Dispenser	Provides chilled water without opening the refrigerator door.

4. Method

4.1 Market Research Methodology

Market research was conducted using a mixed-methods approach that combined questionnaire-based field research with quantitative analysis of product and market data

collected from major e-commerce and price-comparison platforms, including Digikala and Torob. The questionnaire was administered to a diverse group of respondents to identify consumer preferences and requirements, while the analysis of online product and market data was used to examine prevailing product characteristics, price levels, and market offerings. The integration of these data sources provided a comprehensive basis for identifying the

characteristics of the target market and informing subsequent product and innovation strategy decisions. Based on the findings of the market research, the target market was defined as households with 1–5 members, with a particular focus on lower-income consumer segments. Within this market, product positioning was considered in terms of achieving a strategic balance between cost competitiveness and energy efficiency. Accordingly, household size and annual household income were selected as the primary market-segmentation variables and were subsequently used to characterize the target market and guide the analysis presented in Section 5.1.

4.2 Demand Forecasting Methodology

The demand forecasting framework combines quantitative analysis with qualitative assessment of market conditions. The qualitative component considers macroeconomic indicators, import-substitution potential, export opportunities, and international market benchmarks to provide contextual insights into future demand conditions. For the quantitative analysis, given the periodic but non-seasonal pattern of product demand, the Simple Moving Average (SMA) and Exponential Smoothing methods were employed to estimate future demand based on historical observations (Hyndman & Athanasopoulos, 2021). Exponential Smoothing assigns greater weight to more recent observations through a smoothing parameter (α), with values ranging from 0 to 1, thereby allowing the forecast to respond to recent changes in demand while reducing the influence of random fluctuations. The resulting demand forecasts are presented and analyzed in Section 5.3.

4.3 Technical Performance Testing and Energy Certification Protocol

To ensure product quality, safety, and compliance with applicable industrial requirements, domestic refrigerators are evaluated through a standardized set of technical performance tests (Nege & Werke, 2024; Xing & Liu, 2025). These tests assess cooling performance under ambient temperatures ranging from 5°C to 25°C and under different loading conditions; humidity control through weight-loss measurements over time; acoustic performance; energy consumption under standardized test conditions; freezer temperature stability and precision; rapid cooling performance, measured by pull-down time; and structural and mechanical integrity through vibration and shock-resistance tests. Collectively, these assessments provide a

comprehensive evaluation of the refrigerator's functional performance, energy efficiency, and mechanical reliability.

Energy certification is conducted under controlled environmental conditions using a dedicated testing chamber equipped with four monitoring stations. The refrigerator is positioned in a designated test location and subjected to steady-state monitoring for 24 hours at an ambient temperature of 25°C and a relative humidity below 75%. The resulting measurements provide the basis for evaluating energy consumption and other certified performance parameters under standardized conditions. This testing approach is broadly consistent with internationally recognized energy-efficiency certification frameworks, including the U.S. ENERGY STAR program, which evaluates certified refrigerators against applicable energy-efficiency requirements using standardized test procedures (U.S. Environmental Protection Agency [EPA], 2014).

Energy-efficiency requirements are also subject to periodic regulatory revision. In the United States, for example, the Department of Energy finalized an update to residential refrigerator and freezer efficiency standards in 2023, with compliance requirements scheduled to be phased in during 2029–2030 (U.S. Department of Energy [DOE], 2023). Such regulatory developments demonstrate that energy-efficiency benchmarks are not fixed requirements but evolving technical and regulatory criteria that manufacturers must continuously monitor and incorporate into product development and positioning decisions (Nege & Werke, 2024; Scott & Walker, 2026) [538٦٤٨٠].

The principal parameters recorded or verified during the certification process include (U.S. Environmental Protection Agency [EPA], 2014) [412٤١٦٠]:

Energy Efficiency Rating: classification of the appliance according to the applicable energy-efficiency scale.

Annual Energy Consumption (AEC): estimated annual electricity consumption in kWh/year, derived from standardized test measurements.

Storage Volume and Configuration: total usable storage capacity, expressed in liters, together with the relevant internal configuration.

Freezer Capacity: freezer storage volume and, where applicable, the corresponding star-rating classification.

Refrigerator Capacity: usable food-storage volume of the refrigerator compartment.

Frost-Free Status: identification of whether the appliance incorporates a frost-free defrosting system.

Device Identification: standardized documentation of the manufacturer, model, and relevant technical specifications.

4.4 Manufacturing Process

The manufacturing process was documented using process-planning and work-list specification sheets covering fourteen sequential operations, from steel-coil cutting and plastic injection molding to compressor installation, refrigerant piping, and final assembly of the main components (Table 3). The standardized and repetitive nature of this production sequence reflects a product-oriented manufacturing layout built around a relatively stable product architecture. As discussed in Section 6.4, this

stability supports the suitability of an incremental innovation strategy, as product improvements can be introduced within the existing manufacturing structure without major process reconfiguration.

Table 3. Work-list specification: refrigerator assembly operations.

No.	Operation	Description	Main Machine / Tool
1	Door and cabinet body production	Dividing and cutting steel sheets into the required parts — front, top, and side sections.	Steel coil processing
2	Plastic component production	Converting polyethylene material using vacuum-forming machines with matching molds — interior parts.	Vacuum-forming machine
3	Middle section / insulation production	Injecting foam / polyurethane between the body and plastic components — middle and interior section.	Foam injection equipment
4	Compressor production / installation	Prepared in-house or imported — rear, lower section.	Auxiliary parts
5	Refrigerant gas piping	Connecting condenser gas piping throughout the refrigerator — lower and upper sections.	Auxiliary parts
6	Evaporator / cooling unit installation	Bending and installing copper refrigerant piping at the rear of the refrigerator.	Auxiliary parts
7	Thermostat installation	Regulating and controlling internal temperature — upper body section.	Auxiliary parts
8	PCB installation	Located near the compressor / ice-making section, rear or side wall, or top.	Auxiliary parts
9	Condenser installation	Generating cooling and dissipating heat through copper refrigerant piping — body section.	Auxiliary parts
10	Expansion valve / capillary tube installation	Refrigerant gas passes through this component; located at the end of the condenser, after the filter-drier.	Auxiliary parts
11	Filter-drier installation	Absorbs moisture before the refrigerant enters the compressor; rear, upper-right side.	Auxiliary parts
12	Starter relay installation	Supplies power to the compressor when cooling is required.	Auxiliary parts
13	Cooling fan unit installation	Expels heat from the refrigerator's main components; near the rear section and compressor.	Auxiliary parts

5. Results

5.1 Household Demographics and Income Segmentation

Households consisting of four to five members constituted the largest segment of the surveyed population, while low- and lower-middle-income households accounted for the majority of the sample. This distribution is consistent with the study's definition of the target market as small- to medium-sized lower-income households.

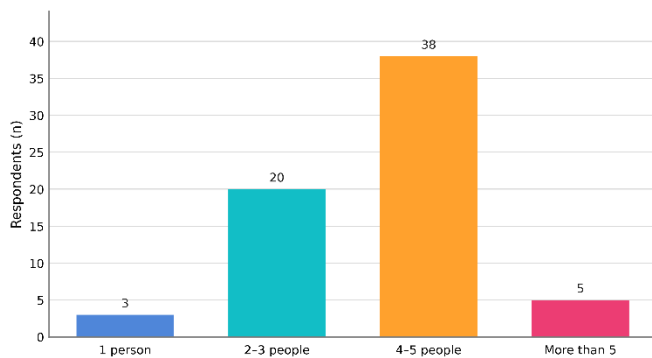


Figure 2. Household size distribution among surveyed consumers.

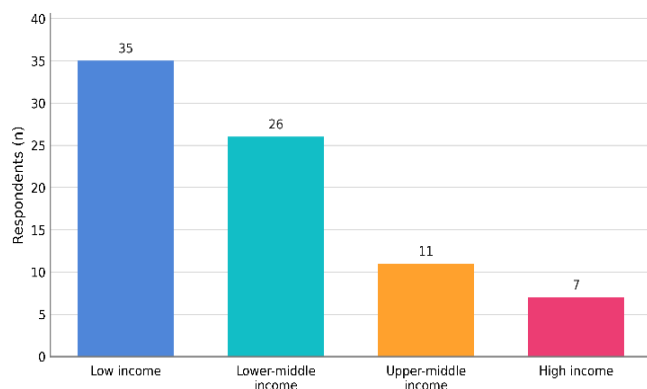


Figure 5. Annual household income distribution among surveyed consumers

5.2 Purchase Decision Factors and Brand Preference

Among the factors influencing consumer purchase decisions, useful life and price rank highest, followed by refrigerator motor quality and external appearance; suitable temperature and payment terms rank lowest. When asked about brand origin, a majority of respondents expressed a preference for foreign brands

over Iranian brands, and among foreign brands, Samsung and LG were the most frequently preferred, followed by Bosch. This pattern is consistent with country-of-origin research showing that a brand's country of origin shapes consumer-perceived quality and brand equity independent of the product's objective attributes (Pappu et al, 2006).

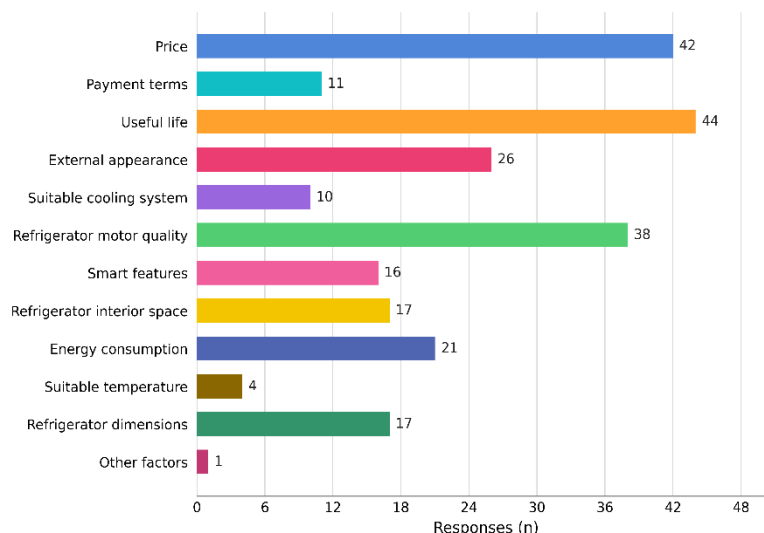


Figure 3. Key factors influencing consumer purchase decisions.

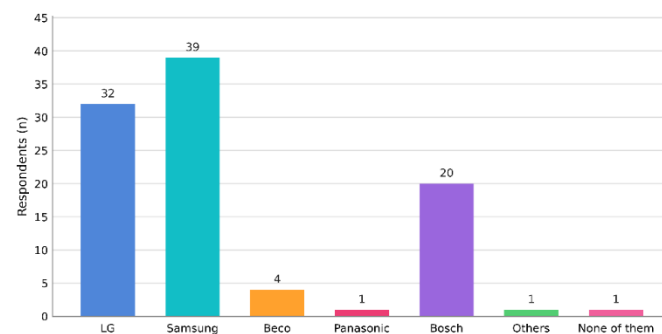


Figure 4. Consumer for Iranian versus foreign brands.

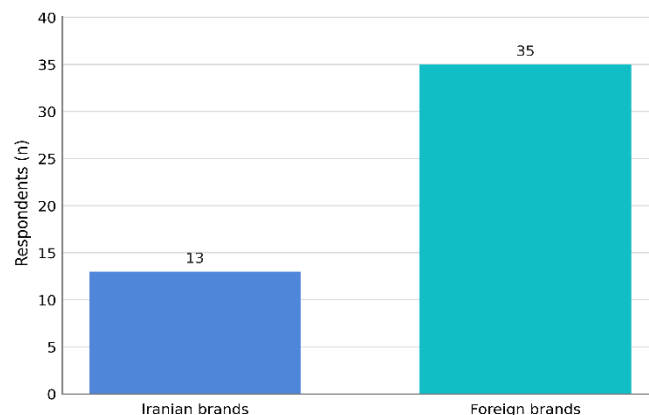


Figure 6. Preferred foreign refrigerator brands among surveyed consumers.

5.3 Demand Forecast

Applying the Simple Moving Average model to the historical sales data (Table 4), the projected demand for 2024 is estimated at 533,824 units, with an average forecast error of approximately 168,279 units. The Exponential Smoothing method, applied to better account for potential cyclical variation, projects demand of 488,048 units for the upcoming year, with a corresponding forecast error of 230,413 units.

Table 4. Simple Moving Average sales forecast.

Year	Sales Volume	SMA Forecast	Actual – Forecast
2019	100,735	—	—
2020	236,791	—	—
2021	464,386	168,763	295,623
2022	551,303	350,589	200,714.5
2023	516,345	507,845	8,500.5
2024 (proj.)	—	533,824	168,279.3 (avg. error)

6. Discussion

This discussion draws together the market, technical, and manufacturing findings presented in Section 5 and interprets them through the theoretical frameworks summarized in Table 5.

Table 5. Theoretical frameworks applied in this study.

Framework	Source	Role in This Study
Technology–Market Innovation Matrix	Strategic Management, Virginia Tech Publishing (n.d., Ch. 7.4)	Classifies the domestic refrigerator's innovation profile along the technology and market axes.

Regular Innovation	Abernathy & Clark (1985)	Explains why incremental, competence-preserving improvement fits the observed production pattern.
Dynamic Model of Process and Product Innovation	Utterback & Abernathy (1975)	Explains the shift from product-architecture experimentation to process- and component-level refinement after a dominant design emerges.
Product Life Cycle Theory	Levitt (1965)	Characterizes the competitive behavior expected once a product reaches the maturity stage of its life cycle.
Appropriability Theory (Complementary Assets)	Teece (1986)	Explains why competitive advantage in this market rests on service, distribution, and warranty assets rather than on the underlying technology.
Ansoff Growth Matrix	Ansoff (1957)	Classifies the appropriate growth strategy (market penetration) for the product/market combination.
Product-Oriented Layout (Operations Management)	Heizer, Render, & Munson (2020)	Explains why a product-focused manufacturing layout suits high-volume, low-variety production.

6.1 Aspiration versus Purchasing Behavior

The results reveal a clear discrepancy between consumer aspiration and actual purchasing behavior. Although foreign brands are strongly preferred, high price levels combined with limited household purchasing power compel many consumers to opt for domestic alternatives. A further inconsistency is observed in the case of brands such as Snowa and Daewoo: while survey responses indicated relatively high interest, user reviews on major e-commerce platforms (e.g., Digikala and Torob) and social media reflect only moderate to low satisfaction levels. Consequently, for new entrants seeking to penetrate the market and capture a viable share, priority must be given to the use of high-quality components and extended product lifespan, while maintaining competitive pricing. In addition, the introduction of flexible financing mechanisms—such as installment plans or bank-facilitated loans—can help reduce economic barriers and accelerate sales volume.

6.2 Competitive Landscape and Market Threats

The domestic market faces intense competition not only from formal foreign brands but also from the pervasive grey market. Significant volumes of unauthorized foreign refrigerators, including new, refurbished, and stock units, enter the country daily. Estimates suggest that a single smuggling operation can introduce 1,000 to 2,000 units into the market per day, assuming a capacity of 34 units per container, creating a formidable barrier for domestic manufacturers seeking to maintain and expand their market share (based on local industry estimates).

This threat can also be understood through the lens of appropriability. Because refrigeration technology itself is neither proprietary nor difficult to imitate, the primary channel through which a domestic manufacturer can capture returns is control over complementary assets such as authorized distribution, warranty coverage, and after-sales service (Langlois & Robertson, 1992; Scott & Walker, 2026; Teece, 1986). The grey market directly undermines this channel by supplying functionally similar products without relying on these complementary assets. This is precisely why the competitive response outlined in Section 6.1 should focus on service quality and financing rather than on the underlying technology (Nege & Werke, 2024; Scott & Walker, 2026). This perspective is consistent with recent evidence showing that, even as digitalization reshapes how firms compete, the ability to appropriate value from innovation continues to depend on complementary organizational capabilities rather than on the underlying

technology alone (Colfer & Baldwin, 2010; Dong et al., 2025; Xing & Liu, 2025).

6.3 Demand Dynamics and Seasonality

Demand for refrigerator freezers is relatively inelastic because refrigerators are essential household appliances (Hertzmänn, 2016; Nege & Werke, 2024; Scott & Walker, 2026). Demand remains relatively stable throughout the year, although it may vary across regions depending on climatic conditions and cooling requirements (Mehdipour & Namazian, 2012). Nevertheless, demand is influenced by two main categories of external factors. Economic factors, including pricing strategies and the timing of sales promotions, can affect purchasing decisions (Nege & Werke, 2024; Scott & Walker, 2026), while social factors, such as the traditional marriage season and Jahiziyeh purchases, referring to household goods purchased as part of a bride's trousseau, can generate periodic increases in demand (as indicated by local market survey data).

6.4 Production Technology and Its Link to Innovation

Domestic refrigerators do not show significant seasonal variation in demand, and the core design does not need to be reinvented annually; manufacturers instead periodically introduce incremental options or minor modifications to the existing platform (Nege & Werke, 2024; Scott & Walker, 2026). Because production volume is high while product variety is comparatively low, a product-oriented (product-focused) layout is, from an operations-management standpoint, better suited to this industry than a process-oriented layout (Langlois & Robertson, 1992; Xing & Liu, 2025). According to the operations-management literature, a product-oriented layout is adopted to achieve the best utilization of personnel and equipment in repetitive or continuous production, where a single product or a family of closely related, high-volume, low-variety products is manufactured (Heizer et al., 2020). This arrangement enables process standardization, lowers unit production cost, and increases line throughput characteristics that correspond closely to the manufacturing conditions observed in domestic refrigerator production, including at firms such as Snowa (a subsidiary of the Entekhab Industrial Group).

From the standpoint of innovation theory, this pattern, preserving the core production technology and product architecture while applying continuous, incremental improvements corresponds closely to what Abernathy and Clark (1985) term "regular innovation." Regular innovation builds on already-established technical and production competence and is directed at existing markets and customers; its effect is to entrench and deepen an

organization's existing skills and resources, and although it often occurs almost invisibly, it can have a substantial cumulative impact on product cost, reliability, and performance (Abernathy & Clark, 1985; Nege & Werke, 2024). This also explains why technical refinement in this industry manifests primarily at the level of manufacturing process and component quality rather than product architecture: once a dominant design has emerged, competitive innovation activity characteristically shifts from product experimentation toward process- and component-level refinement (Scott & Walker, 2026; Utterback & Abernathy, 1975).

This same pattern can be explained through the technology–market innovation matrix widely used in strategic-management texts. When an innovation uses existing technology to improve a product or service aimed at an existing market, it is classified as "incremental innovation": both the technology and market axes remain in the "existing" state, and only the product's features or optional configurations change (Strategic Management, Virginia Tech Publishing, n.d., Ch. 7.4). This aligns closely with the observed behavior of the domestic refrigerator industry, where product refinements occur gradually and within the boundaries of existing technology and an existing market. This classification is also consistent with recent technology-management research on the home appliance sector. Morovati Sharifabadi, Ziaecian, Mirfakhradini, and Zanjirchi (2024) find that home appliance manufacturers are pursuing Industry 4.0 adoption incrementally, layering digital and automation capabilities onto established production platforms rather than replacing them outright, the same pattern of technology-preserving, capability-deepening change identified here. Consistent with this, Talaie, Ziaecian, and Malekinejad (2024), studying quality 4.0 deployment specifically within Iran's home appliance industry, report that manufacturers face significant organizational and infrastructural constraints that favor gradual, phased technology upgrades over radical transformation. The present survey data reinforce this picture from the demand side. This analysis also connects to the Ansoff Matrix. Under Ansoff's model, when an organization continues to operate with an existing product in an existing market and seeks only to grow market share or reinforce its position through incremental product improvements, its strategy is classified under the "market penetration" quadrant the strategy

side: among the purchase-decision factors in Figure 3, "smart features" ranked well below fundamentals such as useful life, price, and motor quality, suggesting that Iranian consumers have not yet made connected or IoT-enabled functionality a primary purchase criterion. Taken together with broader evidence that digital transformation contributes to appliance-manufacturer performance mainly through incremental, dual-innovation pathways rather than wholesale disruption (Xing & Liu, 2025), these findings support treating smart/connected features as a candidate for future incremental upgrades within the existing technology–market cell, rather than as grounds for reclassifying the domestic refrigerator as a radical-innovation or new-market opportunity. This positioning also aligns with classic product life-cycle theory. A product in the maturity stage of its life cycle is characterized by slowing primary-demand growth, intensifying price- and service-based competition among existing players, and a competitive emphasis on incremental differentiation and cost efficiency rather than the market-creating experimentation typical of a product's introduction and growth stages (Levitt, 1965), a description that maps closely onto the largely inelastic, replacement-driven domestic refrigerator market described in Section 5. This account is echoed in recent product life-cycle scholarship, which finds that firms sustaining competitiveness at the maturity stage rely on incremental innovation, market segmentation, and service- or brand-based differentiation rather than fundamental product reinvention (Nege & Werke, 2024).

Technology–Market Innovation Matrix

	Existing Market	New Market
Existing Technology	Incremental Innovation ,Domestic refrigerator (this study)	Market Innovation , New market, existing technology
New Technology	Technological Innovation, New technology, existing market	Radical Innovation , New technology, new market

Figure 7. Technology–market innovation matrix, showing the domestic refrigerator positioned in the existing-technology / existing-market (incremental innovation) cell. Adapted from Strategic Management, Virginia Tech Publishing (n.d., Ch. 7.4).

associated with the lowest level of risk and investment among Ansoff's four growth strategies: product development, market development, diversification, and market penetration (Ansoff, 1957).

Ansoff Growth Matrix

	Existing Market	New Market
Existing Product	Market Penetration — Domestic refrigerator (this study), lowest risk	Market Development — Existing product, new market
New Product	Product Development — New product, existing market	Diversification — New product, new market, highest risk

Figure 8. Ansoff growth matrix, showing the domestic refrigerator positioned in the existing-product / existing-market (market penetration) quadrant. Adapted from Ansoff (1957, pp. 113–124).

Taken together, the analysis of production technology in the domestic refrigerator industry, characterized by relatively stable demand, high production volume, low product variety, and gradual product refinement, provides the theoretical basis for positioning this product simultaneously in the "existing technology, existing market" cell of the innovation matrix and the "market penetration" quadrant of the Ansoff Matrix (Langlois & Robertson, 1992; Nege & Werke, 2024; Scott & Walker, 2026). The convergence of these analytical tools indicates that, at least under current industry conditions, the appropriate innovation strategy for a product such as the domestic refrigerator is not radical innovation or entry into new markets, but continuous, incremental improvement of product features within the boundaries of existing technology and an existing market (Nege & Werke, 2024; Scott & Walker, 2026; Xing & Liu, 2025).

6.5 Limitations

This study has several limitations. First, the market findings rely on self-reported survey responses and e-commerce platform data (e.g., Digikala, Torob), which may be subject to self-selection and platform-specific bias and may not fully

represent offline or rural purchasing behavior. Second, the scope was deliberately restricted to the two-door refrigerator segment because of the proprietary and confidential nature of multi-door technical data held by manufacturers such as Snowa; the findings therefore may not generalize to twin or side-by-side configurations, for which the survey nonetheless indicated strong latent demand. Third, the demand forecasts carry non-trivial error margins approximately 168,279 units for the Simple Moving Average model and 230,413 units for Exponential Smoothing reflecting the inherent uncertainty of forecasting in a market subject to grey-market volumes that are themselves only estimated (1,000–2,000 units per day). Finally, the grey-market and smuggling estimates are drawn from industry approximation rather than verified customs data, and should be interpreted as indicative rather than precise.

6.6 Future Research Directions

Future research could extend this framework by applying the Kano model and TOPSIS (technique for order preference by similarity to ideal solution) to prioritize specific feature improvements based on customer survey data, building directly on the purchase-decision factors identified in Section 5.2 (e.g., useful life, price, motor quality). Future work could also extend the analysis to the twin and side-by-side segments as manufacturer data becomes available, incorporate verified customs or trade data to refine grey-market estimates, and track how the technology–market innovation matrix and Ansoff Matrix positioning evolve as domestic manufacturers close the quality gap identified in Section 6.1.

7. Conclusion

This study's central contribution is theoretical: by jointly applying the technology–market innovation matrix and the Ansoff growth matrix, and situating both within Abernathy and Clark's (1985) concept of regular innovation and Utterback and Abernathy's (1975) dynamic model of process and product innovation, it shows that a mature, high-volume, low-variety product such as the domestic two-door refrigerator is best understood not as a static engineering artifact but as a product whose competitive dynamics are governed by incremental-innovation and market-penetration logic (Nege & Werke, 2024; Scott & Walker, 2026). Under this framing, the appropriate innovation strategy is continuous refinement within existing technology and an

existing market, rather than radical redesign or entry into new markets a conclusion the technical evidence in Sections 3 and 4 reinforces, and one consistent with classic product life-cycle theory's account of competition in a market's maturity stage (Levitt, 1965; Nege & Werke, 2024; Scott & Walker, 2026). Practically, this positioning translates into a specific competitive response: because the underlying technology confers little differentiation, domestic manufacturers compete primarily through complementary assets, component quality, extended service life, after-sales support, and flexible financing (Teece, 1986), in order to close the persistent gap between consumer aspiration for foreign brands and actual purchasing behavior (Section 6.1), while contending with the structural pressure exerted by the informal grey market (Section 6.2) (Langlois & Robertson, 1992; Scott & Walker, 2026; Xing & Liu, 2025). Within the scope of this study, the strategic positioning of the domestic two-door refrigerator as an incremental-innovation, market-penetration product (Table 5) is sufficiently established; Section 6.6 outlines how this positioning could be refined and tested further.

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